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Measuring gentrification through commercial amenities: The case of Paris

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ABSTRACT

Monitoring neighborhood gentrification remains challenging, as most approaches rely on residential indicators or housing market dynamics. While these capture the residential dimension effectively, commercial transformation accompanying neighborhood upscaling has been under-measured. We introduce the Amenity Configuration Index (ACI), a data-driven method that systematically analyzes commercial amenity patterns to provide a complementary lens for understanding neighborhood gentrification. Using Paris municipal data (2014–2023), we construct the ACI from 180 amenity types across 56,000 neighborhood-year observations through agnostic analysis of co-location patterns without imposing predetermined gentrifying categories. While the ACI measures commercial transformation rather than residential displacement directly, validation demonstrates strong correlation with housing prices, median income, short-term rentings and demographic indicators. This empirical association validates commercial change as a meaningful proxy for broader transformation. Our contribution is twofold. Methodologically, we provide a systematic, agnostic approach quantifying commercial gentrification that complements traditional residential indicators and substantively, we demonstrate that commercial patterns reveal transformation dynamics particularly tourism and visitor impacts that housing data alone may miss. The ACI offers a complementary tool for monitoring neighborhood change, informing targeted interventions through both housing and commercial policy.

1. Introduction

Urban change in large metropolitan areas has intensified over the past decades, raising growing concerns about socio-spatial inequalities, housing affordability, and the transformation of local commercial landscapes. Among these processes, gentrification has received particular attention from urban scholars and policymakers because of its implications for displacement, social stratification, and neighborhood identity (Bunten et al., 2024; Jacobs, 1961; Kitsos et al., 2025). While early research focused primarily on residential change and housing market dynamics, recent work increasingly highlights the central role of consumption spaces and commercial transformation in shaping contemporary urban change as a key dimension of neighborhood transformation (Cocola-Gant, 2015; Hidalgo et al., 2020; Zhang & Li, 2025).

This paper addresses the commercial dimension of neighborhood gentrification which remains under-measured despite its visibility and policy relevance. Gentrification scholarship increasingly recognizes the phenomenon as multidimensional encompassing residential, commercial, cultural and political transformations that are interrelated but distinct (Lees et al., 2008; Florida et al., 2023). The residential dimension captured through housing prices, demographic turnover and displacement rates has been extensively studied and measured using transaction records and census data. In contrast, the commercial dimension, i.e. the transformation of retail and service establishment that both reflects and shapes neighborhood change has lacked systematic and scalable quantitative measurement. Existing approaches rely primarily on qualitative observation (Zukin et al., 2009; Hubbard, 2018) or selective counting of specific gentrifying amenities (e.g. coffee shops, art galleries, organic markets or so called “ABC” type) which risk imposing researchers' preconceptions rather than measuring commercial transformation comprehensively. Moreover, commercial landscapes are increasingly shaped not only by residents but also by visitor flows, tourism intensification and commuter consumption and these dynamics may operate on different timescales and spatial logics than residential displacement (Liu et al., 2026; Sigler & Wachsmuth, 2020). In globally connected cities like Paris, neighborhood commercial character reflects the intersection of neighborhood resident preferences, other city

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resident preferences, tourist demand and Airbnb-driven visitor economies, making commercial patterns particularly informative about transformation that housing data alone may miss or detect with temporal lag.

In this paper we introduce the Amenity Configuration Index (ACI), a data-driven indicator that characterizes neighborhoods based on the relational co-presence of retail and service activities as a complementary indicator that systematically quantifies commercial transformation using methods from economic complexity theory. We do not claim that the ACI directly measures residential displacement. Rather, we argue that commercial amenity profiles provide a complementary dimension of analysis that captures consumption-visible neighborhood change with potentially different temporal dynamics than housing markets. The ACI is particularly sensitive to tourism and visitor flows beyond resident demographics and can support distinct policy tools such as commercial rent control or zoning restriction alongside housing interventions. Consequently, the ACI can reveal patterns not fully explained by neighborhood demographics alone. We propose, rather than predefining which amenities signal gentrification, using data-driven methods to discover which commercial configurations empirically characterize neighborhoods undergoing transformation. This agnostic approach makes no assumptions about which businesses come with gentrification, instead, it learns from the data itself which amenity combinations are distinctive and how neighborhoods' amenity configurations evolve over time. Critically, this measure is centered on neighborhoods as consumption spaces that encompass both residents and non-residents. We thus position it as complementary to, not a replacement for, residential indicators. The value lies not in substituting commercial data for housing market analysis, but in enabling triangulation, when commercial and residential indicators converge, confidence in gentrification assessment increases. When they diverge, the pattern is potentially signaling tourism intensification without residential displacement, early-stage residential change not yet reflected in commercial landscapes, or early-stage commercial change not yet reflected in residential dynamics.

We apply this methodology to the case of Paris between 2014 and 2023 using the BDCom database produced by the Paris Urbanism Agency (APUR), which provides a detailed inventory of commercial activities across the city. Paris offers a particularly relevant context to examine these dynamics given its high urban density, strong tourism economy, and growing tensions around housing affordability and short-term rentals (Hubbard, 2018). The City's intense overlap between residential gentrification and touristification make it an ideal setting to test whether commercial configurations serve as meaningful proxies for broader neighborhood transformation when validated against residential indicators.

Our empirical analysis makes two key contributions. First, we construct the ACI from approximately 180 amenity types across 56,000 neighborhood observations in 4 separate years, revealing substantial spatial and temporal variation in commercial character. The index successfully differentiates neighborhoods with distinct commercial profiles from traditional working-class quarters to tourist-intensive zones to upscale residential areas without imposing predetermined categories. Second, we empirically validate the ACI's association with residential gentrification through random forest regression. Commercial amenity configurations are strongly correlated to housing prices and median income, demonstrating that commercial and residential dimensions of gentrification are tightly coupled in Paris. This empirical coupling validates the ACI's utility as a complementary proxy. Commercial transformation correlates strongly with residential transformation, suggesting that both capture aspects of the same underlying neighborhood change process. Neighborhoods where commercial transformation outpaces residential change (high ACI, stable housing prices) may be experiencing tourism intensification or visitor-driven commercial change without residential displacement and this pattern is particularly relevant for Paris given intense Airbnb activity and high intra-city connectivity. Conversely, neighborhoods where residential indicators change without commercial transformation may be in early gentrification stages or subject to specific housing market dynamics (e.g., investment purchases, social housing conversions). The ACI's value lies precisely in enabling these distinctions when used alongside traditional residential measures, providing a more complete picture of neighborhood transformation than either dimension alone.

The remainder of the paper is structured as follows. Section 2 reviews the literature on commercial gentrification and urban amenity dynamics and outlines the Parisian context. Section 3 introduces the conceptual framework and methodology. Section 4 presents the data leveraged in Section 5's empirical application to Paris. Section 6 discusses the implications of our findings and the limitations of the approach. Section 7 concludes.

2. Literature review

2.1. Gentrification

Gentrification is a phenomenon affecting numerous cities across highly diverse national contexts. It often leads to the displacement of lower-income populations, loss of identity and cultural diversity (Coronado & Luque, 2026). Recent phenomena like touristification in metropolitan cities have also highlighted the vulnerability of housing supply when both residents and tourists struggle for the same space (Diaz-Parra and Jover, 2021). In response to these shifting socio-demographic profiles, the supply of commercial amenities may in turn be reconfigured. Entire neighborhoods can lose their cultural diversity and heritage under intense processes of touristification and transnational gentrification as observed, for instance, in Barcelona (Cocola-Gant & Lopez-Gay, 2020; Sigler & Wachsmuth, 2016, 2020).

Gentrification can also be considered as an endogenous process driven by the evolution of the local employment structures, notably through the growth of higher-skilled occupations associated with higher wages (Liu et al., 2025) or the rising value of time for high-skilled workers, which increases the demand for central locations to minimize commuting costs and triggers a self-reinforcing cycle of endogenous improvement in local amenities (Su, 2022). This dynamic is further sustained by positive neighborhood consumption externalities, where individuals' preference for living near wealthier neighbours leads affluent residents to expand into bordering lower-income districts, bidding up land prices and causing original residents to migrate out in a process of “endogenous gentrification” (Guerrieri et al., 2013).

In response or as a cause of changes in the socio-demographic composition of neighborhoods, the supply of commercial amenities may be reconfigured. What is often referred to as commercial gentrification constitutes a strong marker of neighborhoods or cities undergoing broader gentrification dynamics (Glaeser et al., 2018; Guimaraes, 2025; Hubbard, 2018; Couture and Handbury, 2023). The rise of tourism appears as a strong driver of this form of gentrification (Cocola-Gant, 2015), it involves the substitution of locally oriented businesses—primarily serving residents and everyday needs—by more standardized retail outlets, including franchises, which predominantly cater to non-resident consumers with more homogeneous preferences and tastes. As a result, cities may experience a degradation of their cultural heritage and a loss of diversity, paradoxically undermining the very attributes that often attract tourists in the first place. In this context, land-use conflicts may also emerge when local identity and cultural heritage are particularly affected (Muniz & Shmaryahu-Yeshurun, 2025).

Resident populations may be displaced as property owners increasingly prioritize short- or medium-term rental strategies (Dejean et al., 2025; Liu et al., 2026). Those who remain may face higher prices for everyday consumer goods and/or find that the local commercial and service offer no longer corresponds to their needs (Zhu et al., 2026). Municipal authorities may play an ambivalent role, as tourism-oriented

attractiveness strategies or urban renewal policies can reinforce commercial gentrification dynamics. For example, as observed in Kyoto, when urban planning decisions prioritize the construction of new residential buildings and apartments over the renovation of historic structures, the supply of short-term rentals may expand significantly (Liu et al., 2026).

2.2. The Parisian context

Like Los Angeles, Barcelona, London, Shanghai, Tokyo among others, Paris is one of those global cities that concentrate many of the tensions associated with gentrification. Medieval origins and concentric expansion created distance from gentrified historic cores (Ile de la Cité, Le Marais). Land scarcity and Haussmannian architectural homogeneity facilitate incremental apartment-by-apartment upgrading rather than redevelopment. Gentrification operates through gradual socio-economic filtering within existing fabric. As a global crossroads with dense tourism (Bantman-Masum, 2020; Derval, 2010), Paris combines international attractiveness with financialized housing markets. High-skilled employment and global capital sustain real estate pressure, embedding neighborhood change within transnational investment dynamics. Paris hosts extensive Airbnb activity (Dejean et al., 2025; Freytag & Bauder, 2018). Post-COVID "15-minute city" policy (Moreno, 2024) coincided with short-term rentings market expansion. Substantial social housing and municipal intervention buffer displacement but coexist with market dynamics. Residential-to-tourist conversions amplify housing scarcity and commercial restructuring, making Paris a laboratory for platform capitalism regulation (Bei, 2025; Dejean et al., 2025; Robertson et al., 2023). Paris experiences strong residential gentrification-touristification overlap producing commercial standardization and local identity debates. Several quantitative indexes of commercial gentrification have been proposed, from business-directory indexes (Kosta, 2019) to stage classifications (Yoon & Park, 2018), multi-criteria measures (Chen & Zhang, 2024) and machine learning frameworks (Zeng et al., 2022), but these rely on predetermined categories, indicators, or labeled training neighborhoods that define ex ante what gentrifying commerce looks like (see Section 6.2 for a comparison). Our article proposes a systematic, agnostic index detecting structural commercial reconfigurations. While conventional markers (galleries, boutiques, cafes; Zukin et al., 2016; Suire, 2022) may emerge, alternative configurations can appear endogenously. We leverage Paris Urban Planning Agency (APUR) field survey data (BDCom).

3. Conceptual framework

Our indexes are inspired from the economic complexity literature (Hausmann et al., 2014; Hidalgo et al., 2020; Shtob et al., 2025; Batty, 2000). We detail their conceptual grounding along with their methodology to build their interpretations. We first present a conceptual framework focused on the meaning behind differentiated retail locations, and then explain how our empirical methodology integrates it.

Fig. 1 explains our conceptual framework. It outlines the different relevant components and simplifies how they articulate. Neighborhood identity is a combination of place characteristics, some examples of which are provided in the diagram. Flows of people to and from places are determined by the characteristics of these places, and a number of these people are consumers. The consumers they attract are differentiated in three ways. First, in the characteristics and identities they are attracted to. Second, in their spending power and willingness, that is, in the amount of money they are able and willing to spend on commercial amenities. Third, in their preferences for different types of commercial amenities.

Gentrification is often interpreted through the lens of increased spending power and such a view remains with no doubt reductionist. But we also know since Bourdieu (1987) that economic capital is related to cultural and social capital. To put it differently, a higher level of economic capital tends to come with a higher level of cultural capital and renewed social capital. The arrival of higher-income groups is accompanied by specific cultural dispositions, aesthetic preferences and evolving social network structures. In that respect, commercial amenities are not merely responses to purchasing power, they operate as symbolic infrastructures of distinction. As Zukin (2008) states, the proliferation of art galleries, cafés, bars, and boutique retail functions (so called ABC amenities) are visible markers of neighborhood change, signaling the arrival of new middle-class consumers and the cultural revaluation of urban space. This ABC configuration signals a specific configuration of cultural capital characterized by strong mechanisms of social distinction, however, other amenity configurations may arise, reflecting different combinations of economic, cultural, and social capital (Wu et al., 2025; Yuan et al., 2024). Despite its intrinsically reductionist perspective, the amenity configurations identified by our model reveal transformations in the local structure of capitals rather than income alone and may capture other forms of amenities configuration related to cultural capital and preferences. Our indexes (see below ACI/TCI) are independent of city size and facilitate international comparisons. Indeed, by not treating the ABC structure as a context-invariant marker of commercial gentrification, we can compare cities embedded in distinct historical and institutional environments.

In a nutshell, the commercial amenity's presence depends on the presence of consumers willing to spend money on that commercial amenity. As such, observing the location of a commercial amenity informs us on the consumer base in that location. It reveals consumers that prefer the goods and services that amenity offers, that are attracted by the overall place's characteristics, and most importantly that can sustain the commercial amenity with their spending.

The information derived from observational commercial amenity location data is rich and complex and this complexity makes gaining insight complicated. Our method's objective is to reduce these data into readable information, and to allow them to provide insight to

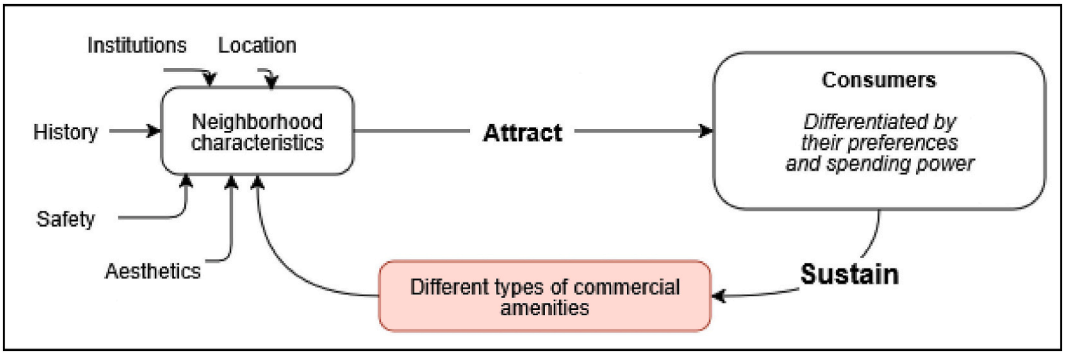


Fig. 1. Conceptual framework of commercial amenity locations in places.

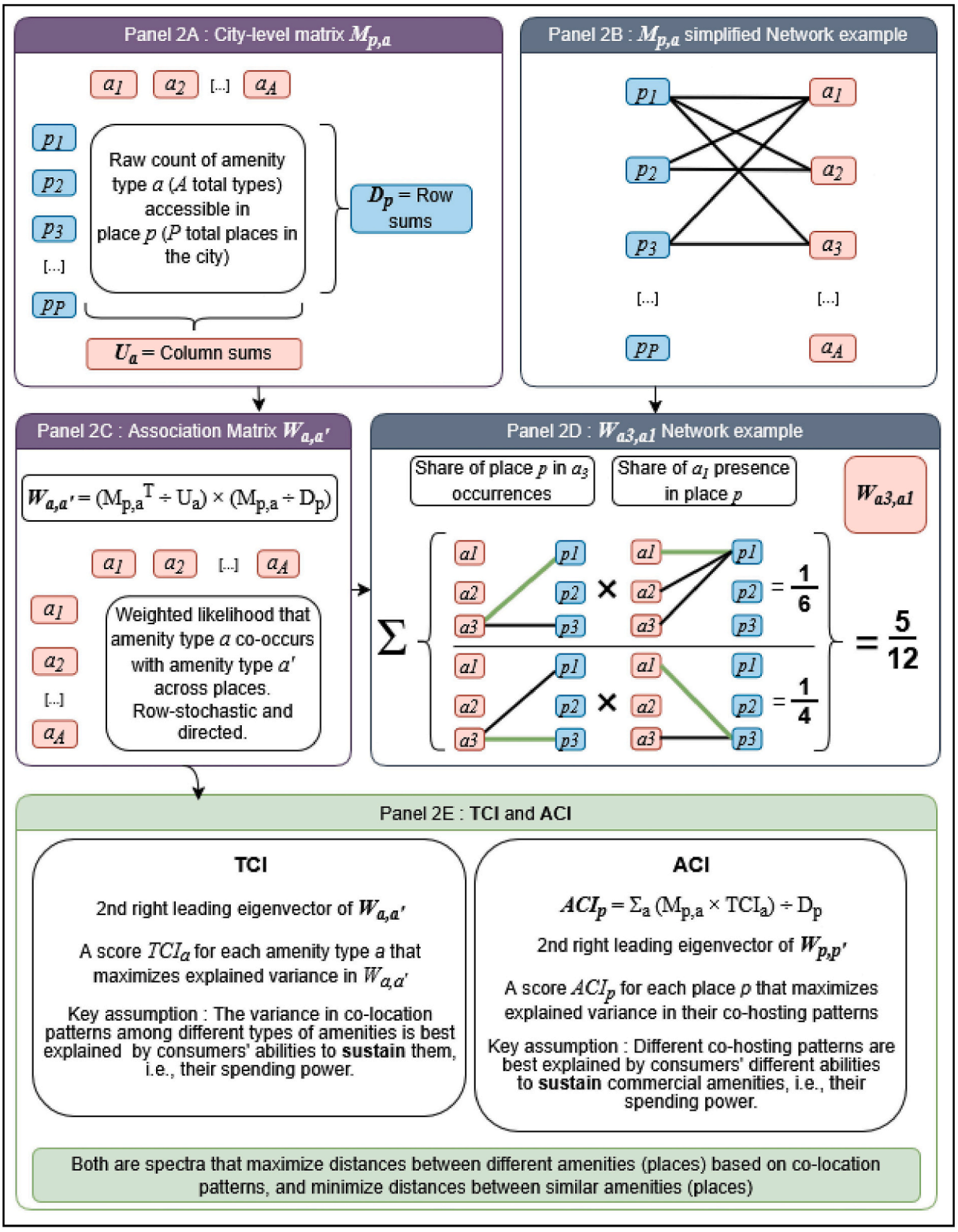


Fig. 2. The Amenity Configuration Index (ACI) and the amenity-Type Configuration Index (TCI) as derived from commercial amenity locations. Panels 2A and 2C present the matrix form explanation of our method, while Panels 2B and 2D illustrate the method on an example network.

researchers and policymakers alike. The first step is to separate commercial amenities into types. In practice, this separation will often be led upstream by the data collector. In this conceptual framework, we consider the best grouping of commercial amenities by type to be attained when commercial amenities of the same type are indistinguishable to consumers, that is, when they are perfectly substitutable goods and services.

Commercial amenities are not the only focus of this analysis. However, they are the only type of amenity we decide to observe. They are easier to observe than most, and as described above, are indirectly representative of the non-commercial amenity landscape of places. However, other types of amenities do not depend on the clear economic condition commercial amenities do. Because non-commercial amenities can sustain losses, and because their locations depend on strictly different conditions (geography, policy), their presence involves even more complex mechanisms. This analysis focuses on a representative feature of cities whose emergence process is made direct by the profitability condition.

3.1. Method

Fig. 2 is where the empirical method to decipher information contained in the locations of commercial amenities is presented, which yields the Amenity Configuration Index (ACI) and the amenity-Type Configuration Index (TCI). Each neighborhood, or place, is paired with each existing type of commercial amenity within a city. They are arranged into a count matrix $M_{p,a}$ where each cell holds the count of commercial amenity type a in place p. A network approach may be helpful to explain the method, and Fig. 2's Panel 2B is an example of a network mirroring $M_{p,a}$. For simplicity's sake, Panel 2B represents a specific case of $M_{p,a}$ in which the count is binary: A type of amenity is either present or it is not. In practice, $M_{p,a}$ would be best represented by a network with weighted edges depending on count. Note that in traditional economic complexity designs, $M_{p,a}$ is binary, not count-based. We deviate from this approach as its reduction further blurs final interpretations and removes information about the intensity of commercial amenity locations.

This method is agnostic in that it does not make any ex-ante assumptions about different types of commercial amenities, especially those, such ABCs, that could be related to gentrification (Zukin, 2008). Instead, it evaluates the relative abilities of commercial amenity types to co-locate. Types of businesses that have a stronger propensity to co-locate are more homogeneous in the consumers they host, thus in those consumers' tastes for both goods and services and places, and in those consumers' abilities and willingness to spend. To expose these relative proximities, we derive a normalized association matrix $W_{a,a'}$ as presented in Panel 2C. Formally, $W_{a,a'}$ is obtained by multiplying the ratio between $M_{p,a}$ and its row sums (the share of amenity type a in place p's total amenity count) with the ratio between the transpose of $M_{p,a}$ and its row sums (the share of place p in amenity type a's total count). This yields a square matrix with A rows and A' columns (for A total amenity types) linking each type of commercial amenity to itself and every other type depending on the relative co-presence with the linked commercial amenity in places. Each cell holds a value between 0 and 1, $W_{a,a'}$ is row-stochastic (row sums are all equal to 1). Note that while the diagonal often holds the strongest value, it is not always the case.

The association matrix is not symmetric and this is by design. Take the example of cinemas and restaurants. In places where cinemas are found, there are often accompanying restaurants for cinema consumers. This would yield a relatively high value in W from cinemas to restaurants. However, take any place where a restaurant is found: It is relatively unlikely to also find a cinema, because restaurants accompany a broad range of other amenity types. Thus, the value in the association matrix W from restaurants to cinemas would be low.

Panel 2D offers an alternative explanation to $W_{a,a'}$ as a normalized similarity matrix and proceeds with the network illustration of $W_{a,a'}$ with the example of $W_{a3,a1}$ as derived from the network in Panel 2B. A random walker perspective can help. Amenity type $a3$ is present in both $p1$ and $p3$, while amenity type $a1$ is present in all of $p1$, $p2$ and $p3$. Place $p1$ is half of $a3$'s presence and $a3$ is a third of $p1$'s present amenity types. A random walker leaving $a3$ and taking two steps on the network would thus have one chance in six of going to $a1$ through $p1$. Similarly, it would have one chance in four of ending on $a1$ having gone through $p3$. The normalized association likelihood from $a3$ to $a1$ can be expressed as the sum of the random walker paths that leave from $a3$ and end in $a1$. It is equal to five twelfths in this example. In formal terms, $W_{a,a'}$ thus functions as a Markov matrix of conditional transition probabilities, and yields the likelihood of ending up on a node a' when leaving from node a. In economic terms, each value in $W_{a,a'}$ is representative of the share of a in the overall landscape of amenities a tends to associate with. It is the relative likelihood of finding a' versus other amenity types in a place in which a is found.

We have noted our deviation from traditional economic complexity methods in using a count-based matrix $M_{p,a}$ instead of a binary matrix (Balland et al., 2022; Hidalgo et al., 2020). It is common practice to binarize $M_{p,a}$ using revealed comparative advantages with a R-star of 1, and an alternative could have been to use a simple presence/absence indicator.

The count-based method does not violate any assumption or mathematical requirement. $W_{a,a'}$ remains row-stochastic and asymmetric, and gains interpretative value. By using it, we avoid threshold effects and arbitrary threshold (R-star) selection. Practitioners that seek to decipher economic insight through the lens of specialization (notably through the diversity and ubiquity they entail) require an RCA-like measure, but such is not the point of our study. We argue that, in practice, the role of these binarization methods (RCA and presence alike) are to inflate the importance of relationships among nodes with scarcer links, and that such inflation is not required in a city-amenity framework. The difference our modification makes in $W_{a,a'}$ is that the relationship between commercial amenities in amenity-dense neighborhoods are not undervalued relative to others. A city can be centralized in its commercial amenity supply by design. We value the preservation of information, and the continuous differentiation between many-to-many, few-to-few or many-to-few co-locations between types of commercial amenities.

The network contained in $W_{a,a'}$ is interesting alone and preserves much of the information in the initial matrix $M_{p,a}$, but it remains complicated to gain macro-scale insight from it given its size. This paper's question of interest is to learn about urban landscapes, demographics and gentrification through commercial amenities. In essence, we ask what constitutes differentiated commercial amenity locations, and what it reveals about differentiated consumer presence. We argue through our association matrix that commercial amenities' relative abilities to co-locate (or not) tell us about how similar they are in the consumers they serve, and in turn about how similar they are in the non-commercial amenities they require to attract these consumers. The next step is to summarize their differences (or similarities) into readable data while keeping in mind our conceptual framework to derive insight.

Spectral clustering tools have been adopted by economic complexity practitioners to reduce these matrices, and we also use them in this framework. Mealy et al. (2019) provide an excellent overview of the method used to derive the economic complexity index and what it implies, and much of their insight can also be applied here.

In practice, the most straightforward method of deriving the amenity-Type Configuration Index (TCI) is to compute the second leading right eigenvector of $W_{a,a'}$, which we call $v^a[2]$. Eigenvectors, when multiplied by their associated eigenvalues, can be candidly seen as decompositions of a matrix that preserve its direction when the eigenvalue transforms it.

As $W_{a,a'}$ is row-stochastic, its highest eigenvalue is 1 and the right eigenvector associated to it $v^a[1]$ is constant. Another possible solution to

this could be to take the left leading eigenvector, which would also be associated with the eigenvalue 1. Recall that $W_{a,a'}$ was normalized by U_a, the column sums of $M_{p,a}$, meaning the first left-leading eigenvector is proportional to the overall presence of commercial amenities types and adds little information.

However, and as the second-best vector at explaining variance in the matrix, $v^a[2]$ is mathematically orthogonal to the left-leading eigenvector (Kemp-Benedict, 2014; Mealy et al., 2019). In other words, $v^a[2]$ is a series of numbers that, once controlled for the total presence (on the global network) of commercial amenity types, best differentiates them. Mealy et al. (2019) demonstrate direct mathematical interpretations. A first network-driven interpretation is that $v^a[2]$ is a partition of commercial amenity types that approximates the minimum normalized-cut value, or criterion. Put simply, it divides observations in network $W_{a,a'}$ into groups in a way that strikes a balance between having low inter-group connectivity (a low sum of weights between the two groups) and high intra-group connectivity (a high sum of weights within the groups).

Another interesting way of understanding this method is through diffusion map distances. $v^a[2]$ solves the problem of assigning numbers to commercial amenity types that minimize the sum of squared distances. The distances that are to be minimized depend on weights in $W_{a,a'}$.

The TCI, $v^a[2]$, altogether offers an efficient way of assigning a single number to each commercial amenity depending on their relative similarities in the association matrix $W_{a,a'}$. It can be thought of as assigning numbers to a one-dimensional readable spectrum that best separates dissimilar commercial amenities from each other, and best groups similar commercial amenities together, based on the commercial amenity types they tend to co-locate with.

The ACI is built on the same initial place-amenity network and leverages $W_{p,p'}$, a place-to-place normalized association matrix analogous to $W_{a,a'}$. The second right-leading eigenvector $v^p[2]$ of $W_{p,p'}$ is the ACI. It best sorts places according to the relative (dis)similarity of the types of amenities they host. Because the ability of a place to host a certain type of amenity depends directly on the presence of consumers willing to sustain that type, the ACI thus separates places by their differentiated present consumers.

3.2. Practical implications

Applying the ACI and TCI method to urban transformations requires careful assessment. Before explaining specifics of this paper's Parisian example, we explain a few key characteristics of the TCI and how to deal with them.

First, it is paramount to understand that $v^a[2]$, the TCI has no natural sign. It is simply a vector that points towards a direction in multi-dimensional space. $v^a[2]$ (the TCI) and $-v^a[2]$ are mathematically equivalent and equally valid. An eigenvector is a vector that satisfies the equation $Av = \lambda v$ (with A a matrix and λ an associated eigenvalue). Because $A(-v) = -A(v) = -(\lambda v)$, $-v$ thus fits the same definition as v with the same associated eigenvalue, and is also a possible eigenvector of A. There is, in fact, no inherent correctness to the orientation of the eigenvector (that is, the relative signs of the values it holds). Different algorithms' implementations can sometimes even yield different eigenvector signs.

This serves to remind the reader that any TCI value alone bears no meaning, as it can only be evaluated relatively to others. But it also poses an analytical challenge. Economic meaning is assigned to the indices by the practitioner, and it is useful to outline the same characteristics on the same side of the continuum consistently through time and space. This is not guaranteed by formulation. Instead of engaging in more complicated formulas, we follow the example offered by Balland (2017) who provides a similar index in his own R software implementation. In practice, after having computed the TCI, we check whether it is negatively correlated to commercial amenity ubiquity (column sums in $M_{p,a}$), and return an inverted TCI if it is not. This ensures consistency across time. We find that it leads to consistent economic underlying meaning.

Likewise, the nature of eigenvectors poses a scaling issue. $v^a[2]$ is only a valid direction, but the scale itself is irrelevant without an associated eigenvector. Cross-year and cross-space comparisons of TCI are thus complicated by nature. The first practical solution to this problem is to use TCI (and subsequently ACI) ranks instead of values where and when possible. Second, in the same way it is standard to normalize values in the Economic Complexity Index literature, TCI values in this paper are standardized (subtracted by their mean and divided by their standard errors). Third, and with practical implication, it is important to consider the temporal scope of the TCI computation. From an economic perspective, rankings may and should be allowed to change through time, as do consumers. To strike a balance between cleaner TCI comparisons and changing real-world conditions we propose a rolling TCI, wherein the eigenvectors that constitute it are built upon a given time's observations along with those directly preceding and following the given time.

Finally, there is a challenge in deriving the Amenity Configuration Index of places from the TCI. Adopting the exact same methodology (with a simple transpose) yields $v^p[2]$, the second right leading eigenvector of a Markov matrix linking places to places $W_{p,p'}$. Depending on how places are defined by the practitioner and on their granularity, this may or may not be computationally feasible. In cases where it is not possible, $v^p[2]$, the ACI of a place, is equivalent to the weighted average of the TCIs of commercial amenities that constitute it (Mealy et al., 2019). This paper's example leverages this property and adapts it, as detailed below. In any case, the interpretative value of the ACI mirrors that of the TCI: It is a spectrum upon which places that are similar in terms of the amenities they host tend to be grouped together, and those that are different tend to be grouped apart. This has interpretative value in terms of non-commercial amenities. But the property of averages brings another TCI-derived interpretation: Places with a higher ACI are places where high-TCI amenities, and thus high-TCI consumers, are found. Differentiating commercial amenities differentiates consumers, and the ACI allows for the spatial identification of these differentiated consumers.

The methods used here make no assumptions about the initial standing or characteristics of commercial amenities, and the results that follow, although impressive given the lack of prior classification, are meaningless without economic intuition. We argue that the leading factor behind differentiated consumption of commercial amenities is consumers' ability and willingness to spend, and that this is thus the characteristic captured by our one-dimensional discrimination method.

3.3. Capturing urban transformations through ACI and TCI

Our approach remains synthetic and accounts for the complexity of cities by being agnostic about the processes behind retail locations. However, the lack of ex-ante classification and assumptions in the ACI and TCI method requires our reading of the indexes to stem from an economic understanding of retail locations. For our indexes to help understand urban situations and change, we must motivate the separation they outline theoretically. Note that this separation is one-dimensional and we are seeking a single leading sorting factor.

Understanding the TCI and the ACI requires us to understand what the one largest segregating factor among consumers is. Although the answer is necessarily intricate, we argue that consumers' spending power is intrinsically linked to their differentiated ability to sustain types of commercial amenities. Hence, the amount consumers spend on commercial amenities determines how they co-locate, and the TCI separates them. As already stated, and because economic capital is correlated with cultural and social capital, changes along the ACI and TCI spectrum reflect broader transformations in neighborhood identity and social composition. What may appear as spending power is in fact the

measurable surface of deeper reconfigurations in capital structures.

In practice, high-TCI amenities tend to be those that co-locate with other selective, leisure-oriented or culturally differentiated amenities. For example, specialty coffee shops, art galleries, organic food stores, wine bars, or boutique fitness studios. Low-TCI amenities, by contrast, tend to co-locate with more ubiquitous and necessity-oriented services such as discount retail, repair shops, basic grocery stores, or administrative services. This distinction is not imposed *ex ante*. It emerges from co-location patterns. As documented by Zukin (2008), if specialty coffee shops systematically appear in places that also host galleries and wine bars but rarely where discount retail dominates, the TCI positions them accordingly: The TCI of coffee shops will be near the TCIs of wine bars and galleries and further away from the TCI of discount retail. The index therefore measures the relative segregation of amenity types along a demand-based dimension.

The Amenity Configuration Index (ACI) translates this logic spatially. A neighborhood whose ACI increases over time is one in which high TCI amenities progressively replace or outnumber low TCI amenities. Empirically, this can correspond to e.g. the replacement of traditional cafés by specialty coffee shops, the substitution of small convenience stores by organic food markets, the emergence of art galleries and concept stores in previously mixed-use areas or the growing density of leisure or tourist-oriented amenities relative to everyday services. Such measurable shifts in amenity composition reflect more than rising income. They indicate that the local consumer base has changed sufficiently to sustain different bundles of amenities under competitive rent constraints. Because commercial space is limited and rents adjust upward in desirable areas, only amenities targeting consumers with sufficient ability and willingness to pay can remain viable.

Thus, when the ACI of a neighborhood moves upward along the spectrum, what is observed is a structural transformation in the local demand composition. In Bourdieusian terms, this reflects a reconfiguration of intertwined forms of economic, cultural, and social capital rather than income alone. Here again, what may appear as an increase in spending power is in fact the observable surface of deeper shifts in habitus, taste hierarchies, and social positioning.[1]

Consumers' residential status (that is, whether they are tourists or non-permanent residents) could be an important determinant because non-residents are likely to consume different amenity types than residents do. However, tourists tend to also come with a lot of spending power allocated to commercial amenities relative to residents for the short time they are present. Separating consumers by spending power will thus tend to separate tourists from residents.

Importantly, the method does not assume that the ABC pattern described by Sharon Zukin is universal. In some cities, high-TCI amenities may include design-oriented services or digital economy-related consumption spaces, in others, they may be tourism-driven. The ACI and TCI framework allows these locally specific configurations to emerge from the data, making it suitable for cross-city and cross-temporal comparisons. In this way, the indices provide a measurable and scalable indicator of commercial gentrification as they detect when and where the composition of urban consumption infrastructures shifts, thereby revealing transformations in neighborhood identity and social stratification. We argue for the ACI and TCI method as a novel way of monitoring gentrification.

We now turn our attention towards the city of Paris to empirically investigate the relationship between local amenity configurations and other place characteristics, and showcase potential descriptive uses of the ACI and TCI method.

[1] The ACI is a relative index, which means it continues to differentiate neighborhoods even at high levels of affluence. Consequently, it can capture processes sometimes described as "super-gentrification".

4. Data

The commercial amenity data we use to demonstrate the ACI and TCI methodology is from APUR's BDCom survey[2] (Paris Urbanism Agency), and it is a strong argument for using Paris as a demonstration. This on-field survey-based documentation of retail and private commercial services in Paris is carried out every 3 years. We have access to data in 2014, 2017, 2020 and 2023, and apply the rolling method mentioned above. Unlike Google Places (used notably in Hidalgo et al., 2020; Juhász et al., 2023; Kaufmann et al., 2022) or OpenStreetMap data, BDCOM is independent of users and of businesses. On top of its dynamic perspective, BDCOM's documentation of market-oriented amenities into categories is more detailed than that of off-field alternatives or of public business registries that tend to group together amenities with poor substitution properties.

The covariate data in the random forest were taken from the DVF database (Housing market price per SQM, publicly accessible), AirDNA (Short-term rental nights spent, private) and INSEE (All other variables, public datasets spanning 2013 to 2021). These covariates correspond to well-established determinants of gentrification processes.

The spatial scope of the overall network and the definition of places are important considerations for practitioners. Our data limits our spatial scope to a study confined in Paris' city boundaries. This spatial scope is not a universal fit for the method, especially within continuous urban areas, but the large ring road that follows Paris' boundaries make for a suitable network in this case.

We propose a definition of places (that is, subdivisions of the city) that respects the urban landscape, provides spatial auto-correlation, and is adaptable elsewhere without the need for administrative boundaries. Here, places as continuous entities through a 15-minute walking cutoff around residential buildings (as defined by the BNDB database), yielding a count matrix linking places to amenity types.

They are built using an implementation (Pereira et al., 2021) of Conveyal's R5 routing engine along with OpenStreetMap street networks. From this count matrix, we derive the similarity matrix of amenity types-to-amenity types upon which the ACI and TCI method is based ($W_{a,a'}$ in Fig. 2). Our cutoff (15 min at 4 km/h) implies that 15 min is an acceptable walking distance to consume, and that any element of the environment in that radius is part of the anticipated location choice of retailers. It appears to be a good compromise for the city of Paris that will be large enough to uncover spatial trends, but small enough that the city is not artificially over-homogenized, and echoes the 15-minute-city literature (Khavarian-Garmsir et al., 2023; Moreno, 2024; Pozoukidou and Chatziyiannaki, 2021).

Paris has roughly 56,000 residential buildings, and BDCOM provides around 180 commercial amenity categories, with minor variations depending on the observed year. This makes the direct derivation of the ACI computationally hard, since a 56,000-row square matrix with large decimal precision is required. We thus leverage the property of averages to compute ACI levels.

[2] "Since 2000, the commercial database, BDCom, has updated an inventory of Parisian formal and legal commercial outlets every three years, using field surveys carried out on all Parisian streets. 9 of such surveys have so far been carried out of which 4 were openly available at the time of writing. The surveyed premises are on the ground floor of buildings, have a display window and visible access giving onto public space. Premises located in courtyards, if accessible, or in shopping malls are also taken into account. Informal commercial activities may occur, especially in tourist or low-income neighborhoods. However, such illegal practices cannot fully substitute for the formal market in Paris where inspections and fines are regularly enforced. The last survey took place in April 2023. A detailed report on developments in commerce in Paris, for each district and for specific themes has been published in 2024". BDCom open data are used by the Paris urban planning agency to produce studies on commercial demography: https://www.apur.org/fr/open-data-cartes/observatoire/observatoire-commerce-paris-bdcom.

5. Results

We present the ACI ranks of Parisian places in Fig. 3 (left). Fig. 3 (right) presents the absolute change in the ACI of places between 2014 and 2023. Places with a low rank are most separated from those with a high rank, and the more yellow a place is on the left panel, the more its consumers tend to be high spenders. Likewise, the more a place is yellow on the right side of the right panel, the more it is trending towards hosting higher spenders and thus susceptible to gentrification.

Spatial centrality is an important factor towards the segregation of places, and thus towards observed differences in present consumers. In an inland city like Paris, being further away from the city's ring road is positively correlated with hosting higher spenders. The importance of spatial centrality is consistent with urban central place theory (Christaller, 1966), with previous attempts to map human activity in cities (Zhong et al., 2017), and with an application of economic complexity to amenities (Juhász et al., 2023). Central locations are however not the only explanatory factor behind commercial amenity separation from a practical standpoint, as Fig. 3 (left) is not homogeneously increasing in rank from a point in the center towards the periphery.

High ranked places in the Center-West (between Annotations 1 and 2 on Fig. 4) tend to be hotspots for traditional tourism and international brands. On top of being well located, they offer aesthetically pleasing architecture (Champs-Élysées), are safer than peripheral alternatives, host museums (Le Louvre: Annotation 2) and monuments (Arc de Triomphe: Annotation 1; Palais Royal, Opéra Garnier), and are close to the river banks of La Seine. This echoes the mechanism outlined in Fig. 1: Non-commercial characteristics attract consumers. If the ACI indeed locates higher spenders, then it should indirectly uncover non-commercial characteristics of places higher spenders prefer. The features of bright places on the map are either universally desirable (centrality) or attractive to high-spending consumers (tourism opportunities, aesthetics). For example, the affluent Parisian West has higher ACI values than the comparatively poorer Parisian East.

However, the role of urban centrality (and of other non-commercial amenities) is less obvious as a feature of ACI change in Fig. 1 (right). Central places have not consistently been getting more different from peripheral places between 2014 and 2023. There is no visible indication of a self-reinforcing process of the ACI in places in the observed period, nor of a polarization of outcomes. Some relatively central places have been hosting higher spenders, but other initially low-ACI places in the periphery have been catching up. Such patterns suggest a commercial amenity composition that is shifting closer towards that of central gentrified places.

A combination of both the initial 2014 ACI rank and ACI evolution is presented in Fig. 4. Here, places have different hues according to their initial ACI rank quartile (brown to blue). This is combined into a two-dimensional color scale with different brightnesses for different quartiles of ACI evolution (dark to bright). The bluer the place, the higher its initial ACI; The brighter the place, the higher its ACI growth.

Cross-year ACI differences do not fundamentally reshuffle the ACI over the city. Urban change is a long process, and rank quartiles remain mostly consistent across years. Still, combined maps such as Fig. 4 remain informative, and the economic trajectories the map reveals are spatially clustered further than the 15-minute neighborhoods that make up places. They are not due to single changes in the amenity composition, but rather to wider spatial phenomena capturing complex gentrification dynamics.

The area East of the centre, denoted by bright colors and affecting all ACI quartiles from lowest (red) to highest (blue), is a fine illustration of what our index can capture. Let us divide it into three areas starting with the neighborhood *Le Marais* (Annotation 4, blue area). *Le Marais* was already described as "the epicentre of bohemian bourgeois French style" by the New York Times in 2014 (Rousselle, 2014). It is one of the first Parisian neighborhoods to have undergone gentrification, notably through state-led initiatives protecting its heritage and restoring its architectural aesthetic. Mermet (2017) finds that what was ongoing gentrification between 1965 and 1990 had already become mature. By the turn of the millennium, it had made *Le Marais* one the city's "most fashionable districts" according to the Telegraph (Gordon, 2005). Despite this, and in large part through tourism, the few leftover fringe streets are still being exploited by gentrifying retailers (Mermet, 2017). Our ACI method identifies this historical trajectory through the initial high ACI of the already-gentrified neighborhood, and it also identifies the still ongoing upscaling process through the ACI growth.

Further North of *Le Marais* (Annotation 3) is the start of the *Canal Saint-Martin* corridor, West of Paris' 11th district. This is an ex-industrial neighborhood that was specialized in textile production until the early 1990s and into wholesale outlets afterwards. It does not have the history nor the aesthetics of *Le Marais*. Yet it is also becoming a trendy neighborhood in part thanks to a public-private partnership (SEMAEST,

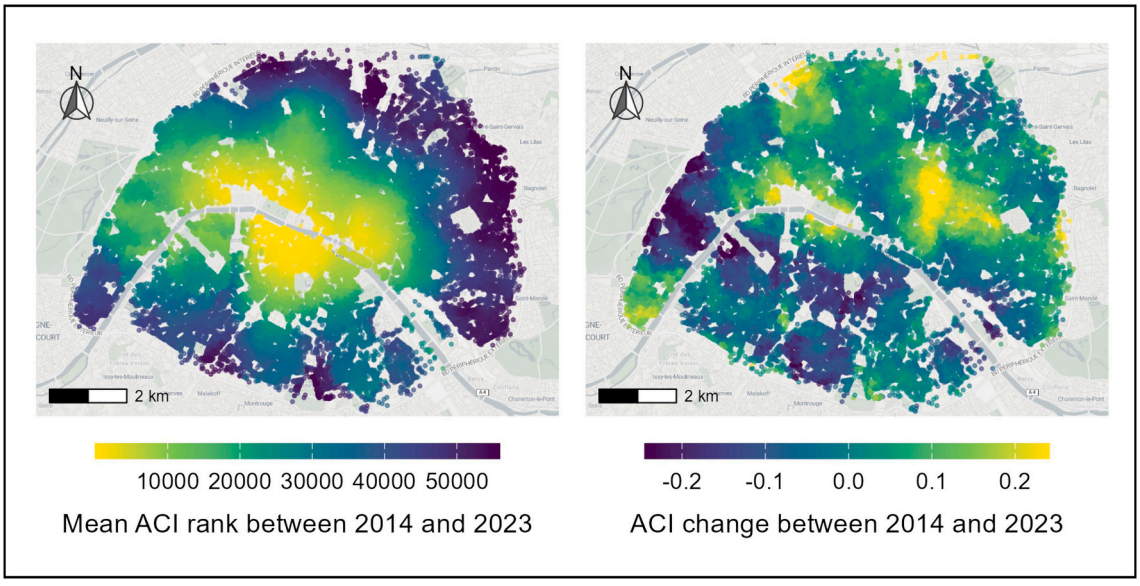


Fig. 3. Central neighborhoods have the highest ACIs, but polarization is limited. [*Left*] Map of the mean ACI rank of 56,000 places in Paris over 2014, 2017, 2020 and 2023. Lower ranks denote higher ACI values. [*Right*] Absolute change in normalized ACI over the observed period (2024–2014).

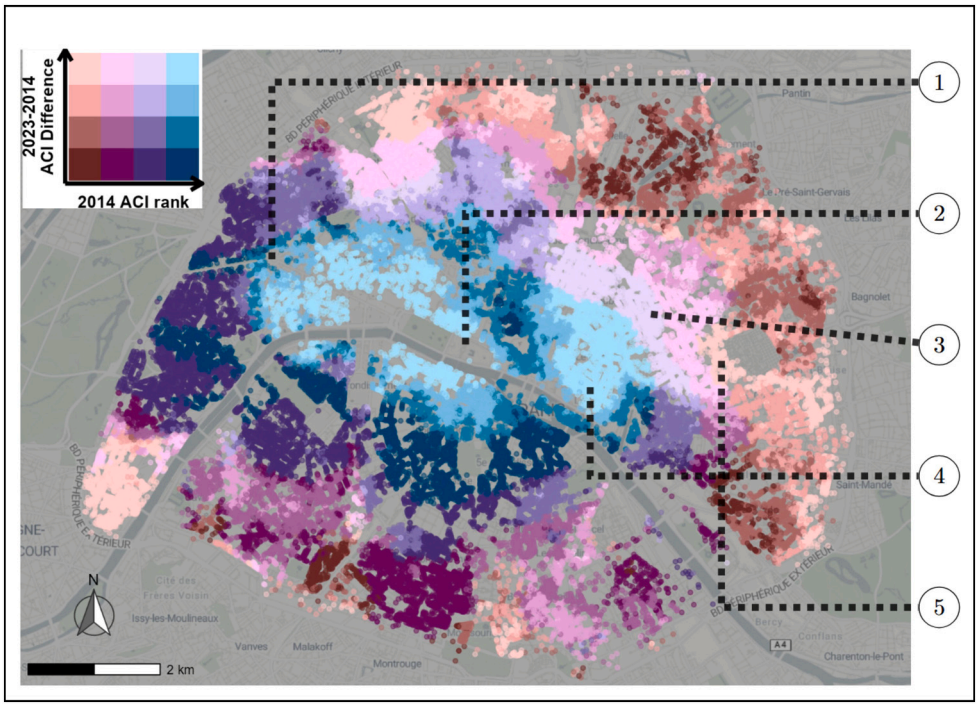


Fig. 4. High-spending consumers are shifting eastwards. These maps combine the initial levels of ACI (top left and bottom in 2014, top right in 2017) and the ACI evolution (top left to 2017, top right to 2020, bottom to 2023) into single maps. Different hues denote different quartiles of initial ACI. Different brightnesses denote different quartiles of ACI change. Mentioned places are annotated and numbered to help the reader.

starting 2004), whose explicit goal in a 2018 revitalisation programme was "never aimed at recreating the pre-existing business mix, but rather at transforming the Chinese wholesale neighbourhood into a trendy consumption hub" (Bantman-Masum, 2020). This is an area where commercial gentrification has well and truly started with coffee shops and ABCs (as evidenced by Bantman-Masum, 2020), but where it has not yet reached the peaks of central Parisian locations. Our ACI Figures are consistent with this analysis: ACI growth is exceptionally high despite its middle-of-the-pack 2014 levels.

Still on that same Eastern side, and perhaps as the most important contribution, the area around Annotation 5 (The West side of the *Père Lachaise* cemetery) is entirely bright too. Clerval (2010) identifies the cimetière as an outpost for Parisian gentrification, but it does not stand out in the literature as a particularly at-risk area. Unlike the two previous examples, the neighborhood hosted types of retail associated with spending levels that were lower than city averages in 2014. Yet, the fast-growing ACI during the observed period outlines the fact higher-spenders are relatively more present in 2023 than they were in 2014. This is a neighborhood where gentrification-induced displacement is particularly likely, be it because of tourism or of knock-on effects of adjacent gentrified neighborhoods.

These three neighborhoods, like any other, are heterogeneous across streets and blocks. But a look at the building closest to Annotation 4's pointer in *Le Marais* reveals that the biggest contributing types of amenities to ACI growth have been Pop-up stores, perfume stores and maintaining Art Galleries. This growth is limited by the addition of fast-food restaurants, which bring down the ACI average.

The ACI growth near Annotation 3 (*Canal Saint-Martin*) is also driven in part by pop-up stores replacing traditional clothing stores and by the opening of new Art Galleries. But the most important factor to this growth, and perhaps the one that best encapsulates our index, is the loss of many in-store telecommunication services (cybercafes, taxiphones) in the area. Taxiphones in Paris are cultural and functional descendents of Senegalese "telecentres", and are an essential part of the commercial fabric for low-income immigrants in Paris. Their removal is a clear indication that these precarious populations are losing social anchors and suggests a renewal of popular neighborhoods to facilitate higher-class appropriation or tourists.

The building nearest to Annotation 5 (*Père Lachaise* West side), ACI growth presents a decidedly different phase of gentrification. The initial 2014 ACI is lower, and ACI growth is most driven by the opening of fast-food restaurants, of fantasy jewelry stores and of tearooms. These amenities are not luxurious or tourist-exclusive, but they are still contributing to retail upscaling in the area.

The Center-West is the traditional tourism hotspot and has only comparatively been losing out to these aforementioned Eastern areas. It has otherwise maintained its status as the highest ACI area of the city. While high-end tourism might be more spread out than it used to be, in part because of the development of short-term rental platforms, the monuments and aesthetics these areas provide still make them attractive to high-spending tourists. However, the affluent residential area West of the tourism hotspot has stagnated, if not comparatively lost out. This map also outlines entrenched disadvantages in areas that are low-ACI and are comparatively shrinking.

Table 1 presents the TCI ranks of the top and bottom 15 Parisian commercial amenities. Lower ranks equate to higher TCIs and vice versa. As detailed in the conceptual framework section, the first level of interpretation of these ranks is a segregation of amenities in their co-location patterns. In other words, the types of commercial amenities presented on the left-hand side of Table 1 are opposites to those on the right-hand side in terms of the other commercial amenities that tend to accompany them in places.

Table 1 is informative regarding the index's robustness and helps cement an interpretation of the TCI separation. Crucially, we observe no large shifts in the rankings across years. The index remains relatively stable and reinforces the idea of a consistent phenomenon being unlocked.

Beyond cross-year consistency, results also appear consistent with expectations regarding which commercial amenities tend to be different from each other. Commercial amenities require sustainment and attract

Table 1
Top 15 (left) and Bottom 15 (right) amenity types by their median TCI rank.

Name	Rank (year)				Name	Rank (year)			
	2014	2017	2020	2023		2014	2017	2020	2023
Luxury general food >300 m^2	1	1	3	16	…				
Haute couture - Designers	2	2	2	2	DIY	170	171	170	168
Tourist hotel - Palace	3	3	1	1	General building work	171	172	171	173
Tourist hotel with 5 stars	4	4	5	4	Moving / Storage	174	173	174	174
Watches	5	5	4	3	Specialty store	179	176	176	175
Art Gallery	7	6	6	5	Driving school	173	175	177	180
Men's shoes	8	7	9	8	Bazaar	178	177	175	177
Antiques	10	9	7	6	Discount store	175	174	181	182
Sale of old books	6	8	10	10	Funeral homes	177	178	178	179
Sale of coins and medals	11	11	8	9	Car composite	176	179	180	181
Generalist Sport	9	10	12	11	African restaurant	180	180	179	178
Department store	13	12	11	7	Telecommunication in store	182	182	183	184
Ice cream shop	12	13	15	15	Discount supermarket	181	181	185	186
Jewelry	14	14	14	13	Hypermarket	183	183	182	183
Women's shoes	15	15	17	18	Specialized supermarket	184	184	184	185
…					Car technical control	185	185	186	187

consumers that prefer them. The presence of given amenities in a place is thus reflective of the presence of consumers willing to consume them, and the separation led in the TCI method separates these consumers.

With that said, it still leaves some space for change. Luxury General Food stores are an apt illustration that may also help the reader understand the index. These commercial amenities are extremely rare, as there were only ever 5 locations of this type on record. La Grande Épicerie de Paris has owned a store located near the Sèvres metro station over the entire time window, surrounded by very high-ACI places. In 2018, they opened a second location near the La Muette metro near the city's Western boundary, surrounded by high but not elite ACI places. The other three locations are from brands Hédiard and Fauchon who were located at *Place de la Madeleine*, at the heart of the city's highest ACI places. As these three locations closed in 2020, the average Luxury General Food store was no longer associated with the high-TCI businesses around *Place de la Madeleine* in 2023. Instead, the 2023 ranking depends only on La Grande Epicerie's locations, which are not as exclusively associated with high-spending businesses.

On the right-hand side, we find goods and services that are either cheap, associated primarily with residential use, and/or with inferior goods. These are most segregated from commercial amenities on the left-hand side, which are strongly associated with tourism, high prices and margins, leisure, and could be qualified as luxury or superior goods. These are typical of gentrification markers (Zukin, 2008). This, in hand with our ACI observations, comforts us in a ACI and TCI interpretation centred around the level of spending of consumers.

Finally, and to cement our index as representative of local realities, we present a random forest regression of Parisian ACI values against sociodemographic characteristics of census tracts often associated with gentrification and urban transformation. We also include short-term rental availability through the Airbnb platform, which has been shown to contribute to gentrification (Dejean et al., 2025; Hidalgo et al., 2023; Basuroy et al., 2020; Ioannides et al., 2019).

We find that the variables presented in Table 2 are strong predictors of ACI levels, as demonstrated by the extremely low prediction error on the out-of-bag data outlined in Table 3.

The relationship between the ACI and these covariates both cement the ACI and provide insights into the drivers of consumption differentiation in space. Note in particular the strong correlations and the covariates' efficiencies in accurately predicting the ACI. This index is based solely on the (co-)occurrences of commercial amenities, yet it captures crucial information about local socio-demographic conditions. These strong associations show that commercial and residential dimensions of gentrification are empirically coupled in Paris. We emphasize that this correlation does not establish causal direction as commercial transformation may lead, follow, or co-evolve with residential change. However, the strength of the association validates the ACI's utility as a complementary proxy that captures neighborhood transformation dynamics alongside residential indicators.

Table 2
Feature importance of different variables within the random forest model, correlations of the same variables to the ACI.

	Feature importance[a] for predicting the ACI	Spearman correlation to ACI
Median income in the preceding year[b]	0.25	0.36
Gini coefficient of income in the preceding year[b]	0.42	0.74
Housing market price per square meter in the preceding two years[c]	1.02	0.86
Short-term rental nights spent from 2016 to 2020	0.23	0.36
Share of 25–39 year olds	0.18	−0.16
Year dummies	0.09	∅

Data for income and SQM prices are inflation-adjusted.
[a] Permutation importance, that is, change in mean squared error (MSE) when the variable is randomly permuted on OOB data.
[b] Except for 2023, where we use 2021 data.
[c] Except for 2014, where we use the two following years.

Housing market price per SQM emerges as the most powerful predictor, which is entirely consistent with the current state of knowledge regarding gentrification. From a purely residential standpoint, places with higher housing prices tend to host higher earning residents who themselves tend to consume higher TCI amenities. This effect should however also be covered in part by the Gini coefficient and median income, both of which have much lower prediction importance. This is because places with higher housing prices tend to exert commercial selection through premise prices: Only high-TCI amenities can sustain the high commercial rents of the Center-West core, and only they have customers willing to pay the price premium to consume there. The inverse is also true, as actual use and foot traffic of high-end amenities may themselves be capitalized into property values.

Perhaps more revealing is the disparity between the ACI's association with Median Income and with the Gini coefficient. High-status commercial landscapes are not just driven by the median wealth of a census tract, but also by inequalities in the distribution of income. Paris as a whole is a rich city, and a high Gini will more often than not be indicative of a relative concentration of ultra-high incomes. This concentration mandates a specific mix of high-TCI superior goods and leisure service that are reflected in the ACI. Moreover, it echoes broader ideas of gentrification, wherein gentrifiers are attracted by authentic places and mixed environments.

Table 3
Model characteristics and feature Importance of different variables within the random forest model (using the full data), correlations of the same variables to the ACI.

Total observations	Training split	Number of trees	Max depth	Out-of–box (OOB) Prediction error (MSE)	OOB R-squared	R-squared on test data
213,622	80%	500	∅	0.7%	0.99	0.99

The role of short-term rentals reinforces a transnational and tourism gentrification narrative. Tourists have relatively high spending power, and tend to be attracted by places with desirable non-commercial amenities. It comes as no surprise that their spatial coverage is positively associated with higher-TCI commercial amenities both as a cause and as a factor and perhaps in neighborhoods where local median income might not otherwise support them.

Finally, the negative correlation between the ACI and the share of 25–39-year-olds is weaker, but it highlights a temporal distinction. Young professionals may well be drivers of growth in the Eastern "bright spots" (the pioneer frontier), but they are comparatively less present in the established, highest-ACI quarters of the Center-West. In mature gentrified areas, these populations are often priced out by older or non-resident high spenders.

6. Discussion

This paper contributes to the literature on gentrification by addressing a persistent empirical gap. Existing approaches typically rely on socio-demographic change, housing price dynamics, or qualitative accounts of commercial transformation. While these perspectives are well established, the commercial dimension of gentrification though frequently discussed remains comparatively under-measured in a systematic and data-driven way. We introduce the Amenity Configuration Index (ACI) as a complementary indicator that systematically quantifies commercial transformation, providing a lens on neighborhood change that complements traditional residential measures.

6.1. Temporal dynamics and causal ambiguity

A critical methodological question concerns temporal sequencing. Indeed, does commercial transformation lead, follow, or co-evolve with residential gentrification? Our data constraints, annual observations over nine years provide insufficient temporal resolution to establish definitive lead-lag relationships, and causal mechanisms likely vary by neighborhood context. We acknowledge this limitation explicitly. But three plausible scenarios are consistent with our empirical findings.

Commercial led gentrification occurs when new high TCI amenities (e.g. art galleries, coffee shops, organic grocery stores, wine bars) signal emerging neighborhood coolness, attracting pioneer gentrifiers whose presence enables further amenity upgrading, eventually triggering housing price increases. These dynamics are strongly in line with what Richard Florida (2002) documents with creative class mobility. Secondly, residential led gentrification occurs when demographic shifts and rising incomes change local demand, causing traditional shops to be replaced by amenities catering to wealthier residents. Co-evolution involves simultaneous commercial and residential transformation through mutually reinforcing dynamics.

Our Paris data show correlation between commercial configurations and housing prices and median income but cannot distinguish these scenarios definitively. All three mechanisms likely operate simultaneously in different neighborhoods or gentrification stages. The *Canal Saint-Martin* corridor may exhibit commercial-led patterns, peripheral areas may show residential-led dynamics while mature gentrified quarters like *Le Marais* may demonstrate co-evolution.

However, the ACI's value as a complementary indicator does not require established causal direction. If commercial transformation leads to residential change, the ACI offers early warning of neighborhoods entering gentrification trajectories. If residential change leads to commercial transformation, the ACI quantifies the commercial manifestation of change already detected through housing data. If co-evolution occurs, the ACI captures the commercial dimension of simultaneous transformation. In all scenarios, combining commercial and residential indicators through triangulation provides more complete understanding than either alone. We position the ACI explicitly as a monitoring tool providing real-time tracking of commercial transformation, with diagnostic capacity distinguishing tourism intensification (rising ACI, stable housing prices) from full gentrification (rising ACI and housing prices) that does not depend on resolving causal ambiguity.

6.2. Methodological positioning relative to existing measures

A growing literature proposes quantitative measures of commercial gentrification and our contribution must be positioned precisely relative to these efforts. A first family of approaches builds indexes from predetermined business categories. Kosta (2019) provides the most direct precedent, using business directories to construct commercial gentrification indexes and codifying three distinct measures, a boutique index, a food index, and an ethnic index. His comparative study of two New York Little Italies shows that the food index best captured diverging neighborhood trajectories. Yet this finding itself illustrates the approach's structural constraint because results depend on which indexes the researcher decides to construct in the first place. Similarly, Yoon and Park (2018) classify commercial gentrification in Seoul into predetermined stages (medium-scale local stores, then small-scale food and beverage, then large-scale clothing chains), assuming a common trajectory across neighborhoods, while Zukin et al. (2009) and Glaeser et al. (2018) track predefined gentrifying categories through business listings or Yelp-type data, an approach recently extended to retail churn as a gentrification signal (Glaeser et al., 2023). All these designs require deciding a priori which businesses signal gentrification, a decision that may not transfer across urban and cultural contexts and that cannot, by construction, reveal unexpected markers.

Recent work moves towards data-driven discovery. Zeng et al. (2022) identify localized gentrification amenities in Shenzhen through supervised gradient boosting on points of interest, revealing context-specific markers, bubble tea shops rather than Western-style cafés. This approach, however, remains supervised because identifying featured amenities requires neighborhoods already labeled as gentrifying, so the discovery is conditional on a prior, typically residential, definition of gentrification. Our method requires no such labels.

A second family develops composite or multi-criteria measures. Chen and Zhang (2024) combine three predetermined indicators (commercial use change, commercial displacement, and commercial rent) through multi-criteria decision-making and GIS overlay for Taipei. Hawkins et al. (2022) construct a composite measure of the gentrification process in Toronto, integrating household income, educational attainment, building permits, and establishment composition within a Bayesian hierarchical spatial framework. This work shares our interest in the coevolution of residential and commercial dimensions. However, these measures either require data rarely available at fine spatial scales (commercial rents) or blend residential and commercial signals rather than isolating a commercial measure, and establishment composition enters as counts of categories rather than as configurations. Relatedly, Chapple et al. (2018) examine linkages between transit-oriented development and commercial gentrification, addressing a specific spatial driver rather than offering a general-purpose index. Finally, qualitative observation (Bantman-Masum, 2020; Hubbard, 2018)

provides rich contextual depth but lacks scalability, while existing agnostic amenity approaches (Hidalgo et al., 2020; Juhász et al., 2023) offer scale but suffer user-reporting bias and lack amenity-specific methodological adaptations for interpreting urban change.

Against this landscape, our contribution can be stated precisely. We do not provide the first quantitative measure of commercial gentrification and valuable indexes already exist. Rather, we provide the first agnostic, configuration-based measure of the commercial dimension, combining four features that no existing index unites. Gentrification markers are discovered from co-location patterns without predetermined categories or labeled training neighborhoods. The relational structure of commercial landscapes is captured through economic complexity methods, rather than through counts of isolated categories. The method requires only georeferenced amenity counts from field-verified administrative data (BDCom), making it scalable wherever comparable data exist. Finally, systematic quantification enables validation against residential indicators. The distinction is not incremental. Where Kosta (2019) asks how intensively predefined gentrifying businesses appear, and Zeng et al. (2022) ask which amenities characterize neighborhoods already known to gentrify, we ask which amenity configurations emerge from the data and what they reveal. This is precisely why context-specific markers such as the disappearance of taxiphones invisible to any boutique, food, or ethnic index and undetectable without labeled neighborhoods become identifiable through our approach. Our measure is moreover complementary to composite frameworks. The ACI could serve as a refined commercial input to designs such as Hawkins et al. (2022), replacing raw establishment counts with a configuration-based signal.

6.3. Empirical interpretation and validation

Building on work that interprets retail landscapes as markers of urban change, our method does not assume predefined categories of "gentrifying" amenities. Instead, the ACI and TCI allow differentiated configurations to emerge endogenously from co-location structures. This addresses an important limitation: the tendency to treat certain retail forms as universally indicative of upgrading, without accounting for context-specific consumption hierarchies. Moreover, approaches relying exclusively on the socio-demographic profile of residents may overlook important dynamics. Consumption spaces are shaped not only by local inhabitants but also by commuters, visitors, and tourists. The interaction between ACI values and Airbnb density highlights how our method identifies tourist-oriented consumption patterns. As a result, shifts in commercial amenities may reflect changes in the socio-demographic profile of actual consumers rather than residents alone. The ACI captures transformations in effective consumption patterns that remain largely invisible when relying solely on residential data.

At the theoretical level, we anchor our interpretation in a multidimensional understanding of capital. Gentrification is frequently reduced to rising income levels, yet economic capital is deeply intertwined with cultural dispositions and social networks (Bourdieu, 1987). Consumption practices are structured by habitus and they signal and reproduce social distinction. Commercial amenities therefore operate simultaneously as economic responses to demand and as infrastructures of symbolic differentiation.

Our empirical findings demonstrate that commercial and residential dimensions are empirically coupled in Paris. The ACI is correlated with housing prices and median income, validating its utility as a complementary proxy for neighborhood transformation. The segregation between leisure-oriented and necessity-oriented amenities is consistent with differentiated spending capacities as a direct sorting mechanism. Yet, the explanatory power of short-term rental intensity and other non-income covariates suggests the mechanism is broader than income alone. The ACI reflects transformations in local demand composition including tourism flows, accessibility changes, and shifts in neighborhood desirability.

When commercial and residential indicators converge (both rising), confidence in gentrification assessment strengthens. When they diverge, the pattern is diagnostic as high ACI with stable housing prices may signal tourism intensification or visitor-driven commercial change without residential displacement and this is a pattern particularly relevant for Paris given intense Airbnb activity. Conversely, rising housing prices with stable ACI may indicate early-stage residential gentrification not yet reflected in commercial landscapes, or specific housing market dynamics (investment purchases, social housing conversions). This triangulation capacity enabled by treating commercial as a complementary dimension rather than replacement for residential indicators provides diagnostic value that neither measure alone can offer.

6.4. Application to others cities

Our Paris focus raises transferability questions. We distinguish methodological framework (transferable) from rankings (context-specific). The agnostic method applies wherever comprehensive commercial data exist, imposing no predetermined categories. Context-specificity is deliberate and capital structures are culturally constructed (Bourdieu, 1987). High cultural capital signals differ across contexts. Comparative evidence supports transferability. Zhu et al. (2026) in China, Liu et al. (2025, 2026) in Kyoto, Muniz and Shmaryahu-Yeshurun (2025) in U.S. Latinx areas validate the methodological principle. Responsible application requires comprehensive local data, meaningful categories and critical interpretation with local expertise.

6.5. Limitations and future research

Several limitations warrant acknowledgment. First, TCI rankings reflect Parisian contexts and require local calibration elsewhere, particularly where informal economies or state retail planning dominate. Second, annual observations limit precise temporal sequencing. Higher-frequency data would strengthen causal inference. Third, focus on formal registered establishments may underrepresent informal commerce. Future research could extend this framework comparatively across cities, integrate temporal diffusion analysis, combine ACI with micro-level business turnover and ethnographic observation, or examine policy intervention effects on commercial transformation trajectories.

7. Conclusion

This paper introduced the Amenity Configuration Index (ACI) and amenity-Type Configuration Index (TCI) as tools to monitor commercial transformation at the neighborhood level. By leveraging comprehensive commercial amenity data and network-based methodology from economic complexity theory, we have provided a tractable framework for capturing differentiated urban demand structures.

Our contribution lies in both measurement and interpretation. The indices are strongly associated with income and housing market dynamics for housing prices, yet should not be understood as simple proxies for wealth. We demonstrate that amenity segregation reflects reconfigurations of intertwined economic, cultural, and social capital. Commercial infrastructures materialize these shifts in ways that are observable, scalable, and comparable across contexts.

For policymakers, this approach offers a complementary lens through which to observe urban change. Because gentrification is a transversal phenomenon affecting housing, commerce, accessibility, and social cohesion, synthetic tools are needed. The ACI provides an amenity-based indicator that can inform planning strategies, retail policy, and equity-oriented interventions. Its diagnostic value through triangulation distinguishing tourism intensification from residential gentrification, identifying neighborhoods experiencing commercial reconfiguration enables more targeted policy responses than residential indicators alone.

While our data cannot establish whether commercial transformation leads or follows residential gentrification, the ACI's demonstrated empirical coupling with residential indicators and its utility as a monitoring tool are robust to this causal uncertainty. By systematically quantifying the commercial dimension of gentrification and enabling triangulation with traditional residential measures, this study contributes to ongoing debates on urban inequality and offers a scalable method to engage empirically with the complexity of contemporary neighborhood change.

CRediT authorship contribution statement

Robertson Calum: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Suire Raphaël:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Dejean Sylvain:** Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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